

1 *Structure of Control Systems*

Draw the block diagram for temperature control in a refrigerator. Use as much detail as you can think of. What disturbances are present in this problem? Clearly label the plant, sensor, reference, and controlled output.

2 *Components of Control Systems*

Consider the problem of holding a car at a constant speed. There is a speedometer that measures how fast the car is moving, cruise control which the user can set, and the motor controls the speed of the wheels. The route runs over a wet, poorly maintained dirt road with potholes.

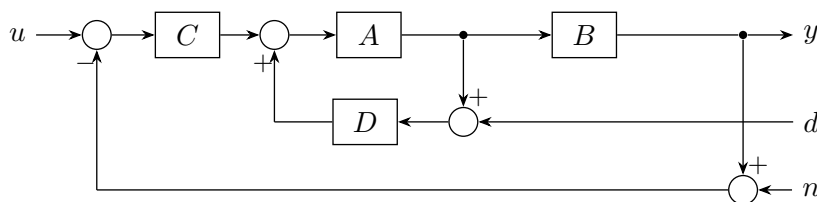
Match the following terms with their appropriate meanings.

- | | |
|--------------------|-------------------------------------|
| (a) speedometer | A. an example of a disturbance |
| (b) mud patch | B. reference |
| (c) cruise control | C. sensor |
| (d) motor | D. another example of a disturbance |
| (e) pothole | E. actuator |

3 *Block diagrams*

Consider the block diagram shown below. Each block is a gain with the letter in the block denoting the numerical value of the gain. Find the matrix M (which is 1×3) such that

$$y(t) = M \begin{bmatrix} u(t) \\ d(t) \\ n(t) \end{bmatrix}$$



4 SIMULINK

Plot the trajectory of a ball thrown up in the air at an angle of 50° with an initial speed of 28 **meters/sec**. The ball has mass 0.25 Kg. Assume that the only force acting on the ball is gravity. Provide a plot of the trajectory of the ball for 5 seconds.

5 SIMULINK *and Modeling*

Now let's throw in wind resistance to the previous problem. Because of wind resistance, the ball experiences a force of $0.028 * v^2$ where v is the speed of the ball. This force is in the direction

opposite to the velocity vector. First derive the equations of motion for the ball. These will be coupled differential equations of the form

$$\ddot{x} = f_1(x, y), \quad \ddot{y} = f_2(x, y)$$

Then, simulate and plot the trajectory of the ball.

6 SIMULINK and Open Loop Control

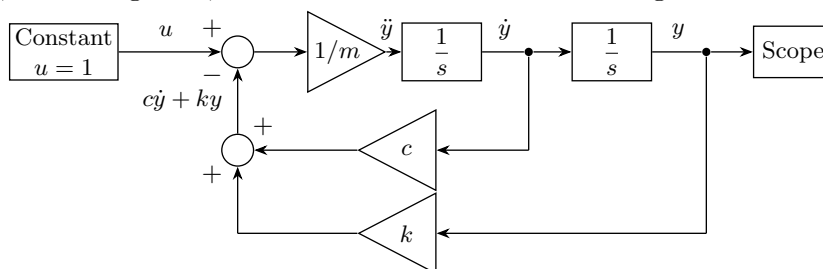
The dynamics for a spring-mass-damper system are

$$m\ddot{y} + c\dot{y} + ky = u$$

where y is the position of the mass, $m = 1 \text{ Kg}$, $c = 1 \frac{\text{Ns}}{\text{m}}$, $k = 1 \frac{\text{N}}{\text{m}}$, and u is the control signal. We wish to choose u to make $y = 1$.

Assume the mass is initially at rest at $y = 0$

Build the model in SIMULINK as shown below: solve the equation of motion for the highest derivative, $\ddot{y} = (u - c\dot{y} - ky)/m$, and integrate twice. The damper and spring terms are formed by Gain blocks, added together, and subtracted from u . Both Integrator initial conditions are 0.



- (a) We use the naive controller that just sets the force to a constant, i.e. $u = 1$. What is the asymptotic position of the mass, i.e compute

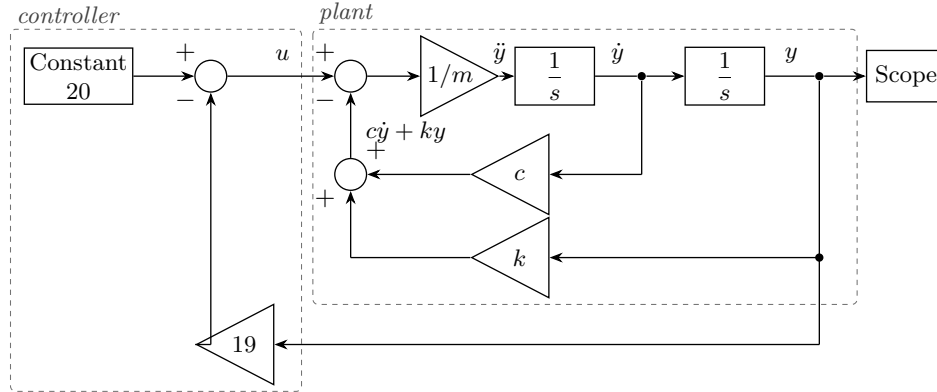
$$\lim_{t \rightarrow \infty} y(t)$$

- (b) Using SIMULINK, compute and plot y versus t for $0 \leq t \leq 12$. Comment on the performance of this controller.
- (c) Suppose there was an error in measuring the spring constant, so the actual spring constant is $\tilde{k} = 1.2 \frac{\text{N}}{\text{m}}$. Use the open-loop controller $u = 1$. Overlay a plot of y versus t for $0 \leq t \leq 12$ on the plot from Part (a) (use the `hold on` command). Comment on the effect of modeling uncertainty on the controller performance.

7 SIMULINK and Closed Loop Control

Consider again the spring-mass-damper system of Problem 6.

The plant is the model you built in Problem 6; only the controller changes. The measured y drives a Gain of 19 that is subtracted from a Constant 20:



- Let $u = 20 - 19y$. Plot y versus t for $0 \leq t \leq 12$. Comment on the performance of this controller.
- Suppose the actual spring constant is $\tilde{k} = 1.2 \frac{\text{N}}{\text{m}}$. Use the feedback controller $u = 20 - 19y$. Overlay a plot of y versus t for $0 \leq t \leq 12$ on the plot from Part (a). Comment on the effect of modeling uncertainty on the controller performance.
- Compare the open-loop and closed-loop control strategies.